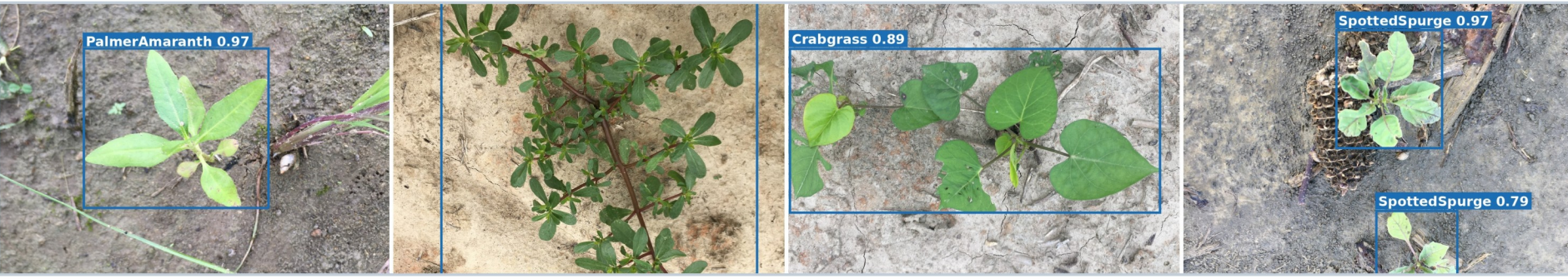


Built at the MTSU Great Robotics Lab. Every robot the lab owns reports into one platform; agents on it harvest, filter, label and train; a third agent watches the other two. This is what runs today, what the robots have recorded, and what six months of running it taught us.

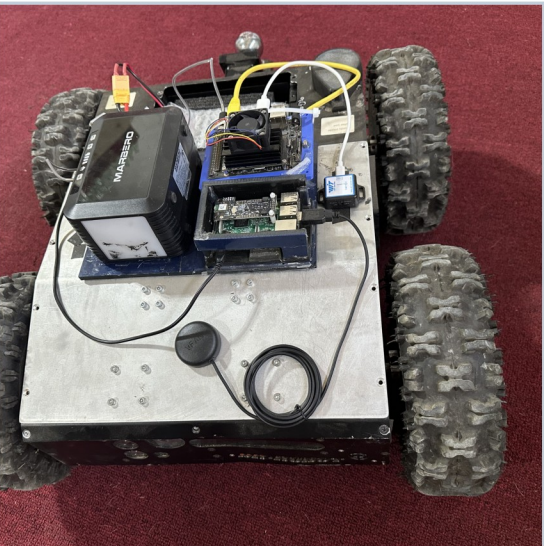
WHAT EACH PROJECT IS, AND WHAT IT IS FOR

Weed Detection



Two agents work the same corpus without being asked. One goes out to public sources, brings weed imagery back and audits what it brought; the other trains a detector on whatever survives the audit and scores it. The registry behind them holds more than 150,000 labelled weed images, and the detector that comes out is small enough to ride on the robots themselves.

241 Robot



A tracked rover built for the row: narrow enough to turn inside one and tough enough to cross the beds between them, so it can reach parts of a field a tractor cannot. It carries a camera, GPS and two inertial units and streams all of them live into the platform. That stream is what an autonomous-driving model for the field is trained on, and a signed-in user drives the rover from a plain browser.

Laser Cart



A twin-laser weeder. A YOLO detector finds the weed in the frame, the two galvanometer lasers are aimed at it and fired, and the weed is killed where it stands -- no chemical, and nothing touches the crop beside it. One camera looks down over the work zone and one looks forward, and both stream into the platform while the cart's own operator drives and fires it.

Humanoid and Quadruped



A humanoid and a quadruped, taken out to the farm to record where wheels do not go. Their project is already open and takes the same ingest contract the rover and the cart use, so the day either of them opens a session it is a governed dataset like any other.

Figure 1. Every project on the platform was opened the same way, sends through the same ingest contract, and is trained and evaluated by the same agents.

HOW A ROUND RUNS, AND WHAT IT DOES WHEN IT IS STUCK



Figure 2. The loop an agent runs to answer a need, drawn as a loop. A need arrives, the agent works out what data would answer it, goes and gets it, trains on what survives the audit and scores it, then reads its own result and decides what to ask for next. When it cannot decide it climbs the ladder inside the loop, each rung reading the one below it and free to overrule it. The outward run is solid because it runs unattended today. The return run is dashed because it is the one edge the platform has not closed: the decision is recorded and nothing reads it back, and closing it is the work this poster is about.

ONE PLATFORM, ANY ROBOT, ANY DATASET

■ agentAI is one platform for every robot the lab owns

A robot, a drone or a sensor rig is a project. It owns its keys, its people and a locked dataset registry, and the agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.

■ Weed detection is one project here, the humanoid is still empty

Open a project for any data type and any field, upload, and get an automatic diagnosis before choosing what to do next. Weed detection is one project here; so are the rover and the cart.

■ Adding the next robot takes a project, a key and a session

A new robot needs a project, an API key and a session start. The ingest contract is the same for every vehicle and the platform does not care what the robot is. Empty projects for a mobile robot and a humanoid already sit on it, waiting for a first upload.

■ Tracked rover and laser cart are the two robots reporting in

Robot 241 is a tracked rover that streams camera, GPS and two IMUs. The laser cart carries a galvanometer laser, a down camera over the work zone and a forward camera.

■ A drive becomes a dataset the moment it opens

A vehicle opens a session and the platform registers it as a governed dataset at that moment, while the drive is still running. Frames and sensor batches arrive over one API key. Robot 241 sends camera, GPS and two IMUs; the laser cart sends camera, laser state, vehicle telemetry and its own detections, and no position at all.

■ Signed-in users drive the rover from a plain browser

A signed-in user opens the 241 project and clicks Drive. The platform reverse-proxies the robot's own console over the lab tailnet, so the rover is driven from a plain browser with no VPN client. 1,389 joystick commands have gone through that path. Two gates sit in front of it: the platform's Google login, then the robot's own password.

A BRAIN THAT GETS STUCK AND ASKS UPWARD

■ Grounded recall separates the reviewers

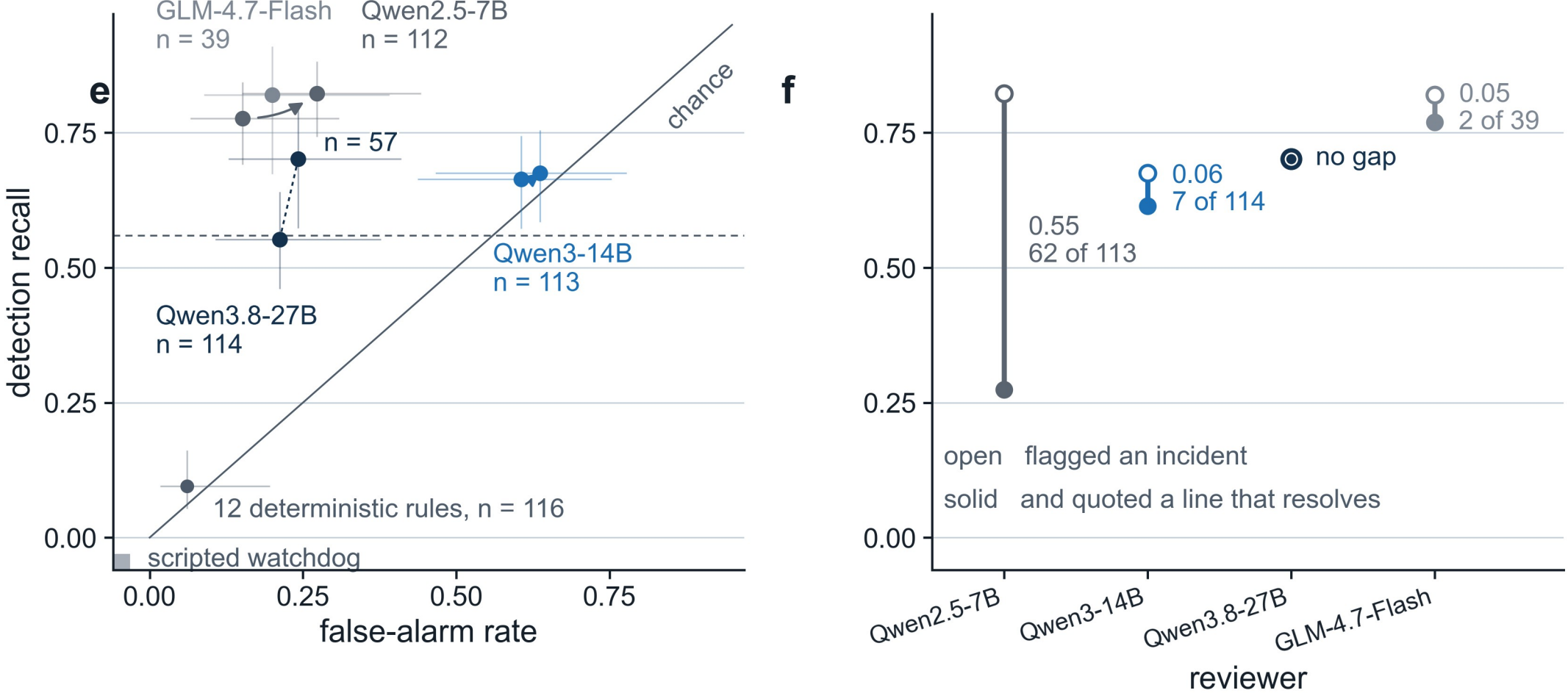


Figure 3. (e) Recall and (f) grounded recall on one frozen corpus of real incidents from this project's own record, scored by the project's own scorer. Bars are 95% Wilson intervals and each mark carries its own denominator. A pair's tail is the model reading raw artifacts and its head is the same model given retrieval over them; the dashed rule is the ceiling a rules-only arm can reach, and an arrow is dotted where its two ends were scored on different cases.

Escalating is worth it, and the reason is grounding rather than size: retrieval is worth +0.149 recall to the 27 B and +0.012 to the 14 B. This part is built and is switched off today.

■ A cheap reviewer, and a bigger one above it when it is stuck

Review runs in tiers: a scripted watchdog, twelve deterministic checks, a model reading the raw artifacts, then the same model given retrieval. On 149 frozen cases the scripted watchdog returned no decision at all.

■ Where a model runs decides whether its verdict counts

A router maps a role -- collector, judge, reviewer, planner -- onto a model and a place, and the place decides whether the verdict counts. Deciding work runs on the cluster, on open weights; the lab box hosts only the small guide.

THE AGENTS RAN THE LOOP WITH NOBODY IN THE ROOM

■ Fifteen rounds ran with nobody in the room

Two cluster jobs, one collecting and one training, ran fifteen rounds against the locked registry with nobody in the room. Every round reported success. This part is built and is switched off today.

■ The corpus stopped growing at round three

The corpus stopped growing at round three: the same 24 datasets went in every time, and the apparent growth was one staging directory nobody cleared.

■ Each finished drive is diagnosed without anyone asking

Every finished drive is diagnosed without being asked -- route, noise, cross-sensor correlation, a timeline -- for 26 live robot sessions and 45 charts so far.

MORE DATA STOPPED HELPING, SO WE WENT AFTER THE LABELS

■ More harvested data cost accuracy, and we measured how much

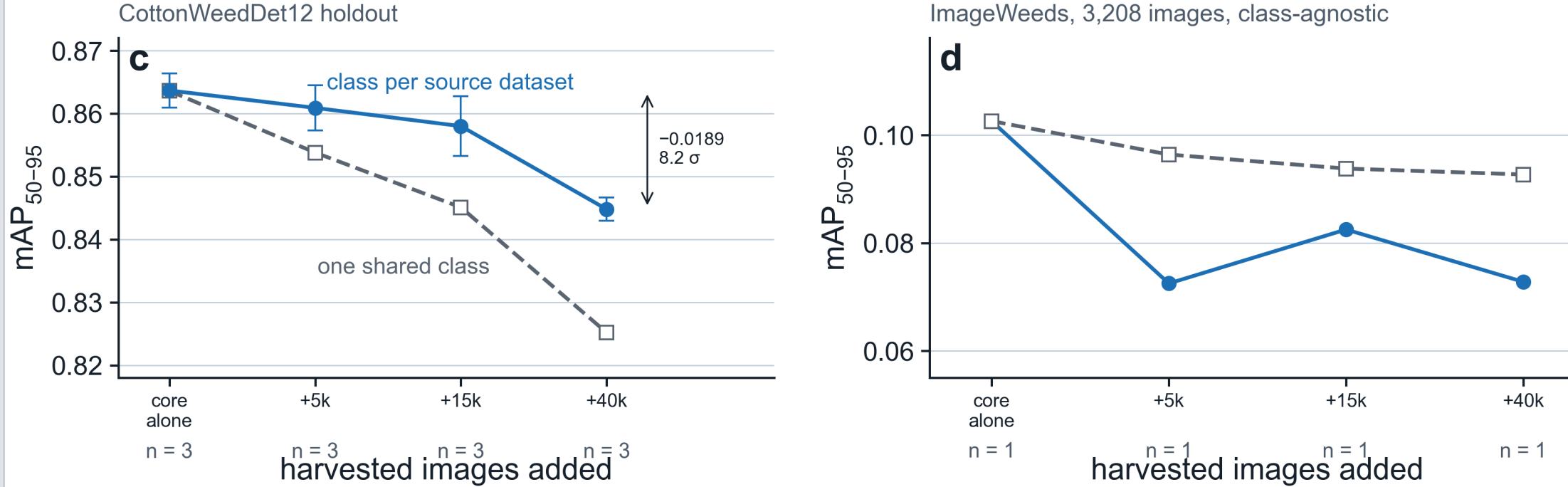


Figure 4. (c) Three seeds at every rung on our own sealed holdout, marks are the mean with 1 s.d. (d) The same eight checkpoints on a second weed dataset they never saw, one seed per rung. Both exams fall.

Starting fresh instead of from the last round's weights recovers +0.0287, 5.1 times this recipe's seed spread. That is the change the next campaign makes.

MORE DATA STOPPED HELPING, SO WE WENT AFTER THE LABELS (CONT.)

■ More data stopped helping, and the training recipe was not the reason

The collector kept adding harvested weed images and the detector kept not improving on them. The training recipe was the first suspect, so the training-side fixes went first: augmentation, replaying the species the model already knew, completing the labels that were missing. Each left the same gap behind, and what the elimination left standing was the data itself -- the boxes our own auto-labeller was writing.

■ A vision model decides whether a dataset belongs, and a chat model cannot

Professor Zhang set the rule: have the system compare what it collected against what we already trust, and discard what sits too far away. A chat model cannot do that job -- asked to draw the box itself Gemma 4 scores 0.012 F1, and shown a cropped candidate box it rejects none of the false ones -- so the comparison is cosine geometry on DINOv2 features, which has no way to rubber-stamp. Trusted sources average 0.78 against 0.26 for untrusted with no overlap, 51 off-domain datasets were flagged out of the corpus on it, and then the cleaned corpus was retrained and the accuracy did not come back.

■ The unit of curation has to match the unit of noise

An auto-label error is one wrong box inside an image that is otherwise fine, so a filter reading a whole dataset cannot see it. The per-box tools are written -- crop every labelled box, score the crop against an object-level reference, and train a head that flags a box whose picture disagrees with the class it claims -- and neither has been re-measured since the bug that made their first run score nothing. That is the open front of this work, and the sheet says so rather than implying a result. This part is built and is switched off today.

■ A brain filed a hypothesis with its own falsification condition, and it was wrong

The harvested images carry no plant label at all: the merge gives every box a class derived from the website the image came from. A brain tier proposed that this bookkeeping was what made the curve fall, wrote the condition that would falsify it before any job ran, and filed it for approval; a human approved it, and four runs falsified it on that condition. What the runs exposed instead is density: nine boxes on a harvested image against under two on a clean one.

WHERE THIS GOES: ONE PLATFORM FOR ANY DATASET, ANY FIELD, ANY ROBOT

The end state is one general platform. Agents on it run without being asked, iterate on their own output, escalate to a larger model when they are out of their depth, and are fenced in a sandbox while they do it. Every capability stays in the platform after the experiment that needed it -- the routing table, the ingest contract, the dataset registry, the sandbox, the scorer -- so the next student inherits them working. That is the difference between a lab that has run an experiment and a lab that has built somewhere to run the next one. This is the direction, not a description of what runs today.